# Aditya Rai

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## Education

**Vellore Institute of Technology (VIT), Vellore** 2023 – 2027
*B.Tech in Information Technology* *Vellore, Tamil Nadu*

**RMVM, Gurugram** 2023
*Class XII (PCM + CS), Score: 81%* *Gurugram, Haryana*

**RMVM, Gurugram** 2021
*Class X (Best of 5), Score: 91.8%* *Gurugram, Haryana*

## Experience

**Trayistats AI Technologies Pvt. Ltd., Noida** May 2025 – Jul 2025
*Software Development Intern* *Noida, Uttar Pradesh*

- Built 6+ responsive pages and reusable components for the Trayistats AI Technologies company website using the MERN stack (MySQL, Express.js, React.js, Node.js), working in 2-week Agile sprints.
- Integrated 5+ REST API endpoints between the React frontend and Express.js backend to load and update website data dynamically, replacing manual content updates.
- Improved the Lighthouse accessibility score from about 75 to 90+ by fixing 5+ accessibility issues such as missing alt text, low color contrast, and unlabeled form fields, and fixed layout issues across mobile, tablet, and desktop screens.

## Projects

**AgentFlow AI – Multi-Agent Market Intelligence Platform** *May 2026*

- Built a 6-agent LangGraph pipeline with FastAPI and Google Gemini that collects live data from YouTube, Amazon, and Reddit, runs VADER sentiment, TF-IDF keyword, and NMF topic analysis, and generates self-reviewed PDF/DOCX reports.
- Implemented RAG with Gemini embeddings in ChromaDB to answer follow-up questions on each report with source citations, rebuilding the index automatically after host restarts.
- Deployed on Vercel, Render, and Neon (FastAPI + PostgreSQL, 30 REST endpoints, JWT, RBAC, 215 CI tests); cut server memory by 42% by replacing scikit-learn with NumPy TF-IDF/NMF.

**QuantPredict – Multi-Model Stock Direction Prediction System** *Jun 2026*

- Developed an end-to-end stock market direction prediction pipeline using historical OHLCV data and 38 engineered features including RSI, MACD, moving averages, Bollinger Bands, ATR, OBV, ADX, rolling statistics, and lag features.
- Trained and compared 6 machine learning classifiers including XGBoost, Random Forest, SVM, and KNN, alongside an LSTM sequence model and Vision Transformer for next-day price movement direction prediction.
- Implemented weighted-voting ensemble learning with time-aware evaluation and walk-forward validation to combine model predictions while preserving chronological ordering and preventing test-period information leakage.

## Achievements & Certifications

- Semi-Finalist, HackBattle Hackathon (200+ teams) – IEEE CS Chapter, VIT; built and pitched a prototype Sep 2024
- MERN Full Stack Certification (98%) – EThnus Codemithra
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – 93%
- SQL and Relational Databases 101 (DB0101EN), provided by IBM Skills Network – 91%

## Leadership & Extracurricular

**Technology and Gaming Club, VIT** Aug 2024 – Apr 2025
*Senior Core – Point of Contact (POC)* *Vellore, Tamil Nadu*

- Managed club events as the Point of Contact and led TAG Clash and OverClock with a 40-member team, handling budgeting, sponsorships, stakeholder coordination, and tournament scheduling.

## Technical Skills

**Languages:** C++, Java, Python, JavaScript (ES6+), SQL, HTML, CSS

**Web / Frameworks:** React, Node.js, Express.js, FastAPI, MERN Stack, TypeScript

**Machine Learning & AI:** scikit-learn, XGBoost, Random Forest, SVM, KNN, LSTM, Vision Transformer, NLP, RAG, Google Gemini, LangGraph

**Data Analysis & Visualization:** pandas, NumPy, Matplotlib, Plotly, Streamlit, yfinance

**Databases:** MySQL, MongoDB, PostgreSQL, ChromaDB

**Tools & DevOps:** Git, GitHub, GitHub Actions, Docker, Docker Compose, Postman, Jupyter Notebook, Vercel, Render